

Dominant Strategy Theory (Game Theory)

"A **Dominant Strategy** is a strategy that gives a player the best outcome **regardless of what the other player does.**"

In simple words:

"No matter what my opponent chooses, this is still my best choice."

If a player has a dominant strategy, they do not need to guess what others will do. They simply choose that strategy.

Example 1: Prisoner's Dilemma

Two criminals are arrested separately.

Prisoner B	Stay Silent	Confess
Prisoner A: Stay Silent	-1, -1	-10, 0
Prisoner A: Confess	0, -10	-5, -5

Analyze Prisoner A

If B stays silent:

- Stay silent = -1 year
- Confess = 0 years (Better Choice)

If B confesses:

- Stay silent = -10 years
- Confess = -5 years (Better Choice)

Confess is better in both cases.

Therefore:

Dominant Strategy for A = Confess

Similarly,

Dominant Strategy for B = Confess

The result (Confess, Confess) is also the **Nash Equilibrium**.

Key Rules

1. Dominant Strategy

A strategy that is always best regardless of what others do.

2. Strictly Dominant Strategy

Always gives a higher payoff.

3. Weakly Dominant Strategy

Never worse and sometimes better.

4. Dominated Strategy

A strategy that is always worse than another strategy and should not be chosen.

Quick Test

You and a friend are choosing whether to carry an umbrella.

Weather	Carry Umbrella	Don't Carry
Rain	Comfortable	Wet
No Rain	Slight inconvenience	Comfortable

Question:

Is "Carry Umbrella" a dominant strategy?

Think about it before answering.

The answer is **No** because if it doesn't rain, not carrying the umbrella is better. Therefore, no dominant strategy exists.

KEY NOTE: Its Important to understand and identify the dominant Strategy.

So how can we do that?.... By asking question and playing the game inside your own mind before you make a decision.

Ask:

If the opponent player chooses option 1, what should I do?

If the opponent chooses option 2, what should I do?

"If my answer is always the same, then it's a dominant Strategy"